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Mesoscopic Traffic Flow Model for Agent-Based Simulation

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Abstract

Traffic simulation is a key element of agent based models as the various agents decisions impact and are impacted by the realized travel times and delays in the traffic network. Here we present the mesoscopic traffic flow model implemented in POLARIS transportation systems simulator. The model is mesoscopic in an attempt to obtain enough for applications that requires microscopic data while still inheriting the computational efficiency of macroscopic traffic flow models. The model is macroscopic at the link-level with dynamics based on the Newells Model and microscopic at the node level. We present link-level and network-level results for one mid-sized traffic network. The results show that the model can combine accuracy, level of details and computational efficiency.

Keywords: Traffic Simulation; Mesoscopic; Link Transmission Model

1. Introduction

The evolving complexity of transportation networks had lead to the integrated modeling of travel demand and network supply as opposed to a sequential process. In particular, Agent-Based-Models (ABM) capture these interactions on the perspective of a particular agent that engage on activities (demand) based on time-dependent travel times for the different modes available (supply) in such a way that demand and supply are mutually dependent. In such tools, the traffic simulation model should be reasonably accurate to provide realistic travel times while computationally efficient so that it can be applied in large scale networks.

In ABM, each agent is represented by multiple models that ultimately define several decisions, from long-term, such as car-ownership, to short-term, such as activity location, departure time, mode, and route for the forthcoming trip.

Two conflicting requirements make the implementation of a suitable model challenging. On one hand, it is desirable to have as much detail as possible so that the outputs are more useful for further decisions of the agents and more

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realistic. For example, estimating fuel consumption requires higher level of detail in vehicle-level trajectories such as speed, acceleration, deceleration, idling etc.. In that regard, microscopic simulation would be a natural choice. On the other hand, the simulation model should be computationally efficient in order to be applicable in large networks. In that case macroscopic models would be preferred over microscopic models as the latter requires detailed network representation, various parameters to be calibrated and is computationally intensive.

Here we present the traffic simulation model developed for the POLARIS tool. The model is mesoscopic in an attempt to obtain enough for applications that requires microscopic data while still inheriting the computational efficiency of macroscopic traffic flow models. The model is macroscopic at the link-level with dynamics based on the Newells Model [7] and microscopic at the node level with flow computation based on the underlying intersection type.

This paper is organized as follows. In Section 2 we present a brief review of related literature and POLARIS. In Section 3 the POLARIS traffic flow model is presented. In Section 4 we present the results of the POLARIS traffic simulation. Finally, we state our conclusions and future work in Section 5.

2. Related Literature

Recently several packages have been developed for integrated agent-based modeling. OPUS [8] is a Python-based open-source library designed to be easily extensible allowing researchers to integrate specific models for their needs such as demand modeling and dynamic traffic assignment. TRANSIMS [6] is an early attempt to integrate different transportation data information, such as land-use and demographic with human decision making. Each agent has a set of plan that is adapted based on the resulting congestion of a given plan.

The concept of plan creation and adaptation is also present in MATSIM [5], an open-source tool for agent-based modeling. Each person is modeled as an agent that adapts its itinerary iteratively based on scores computed based on realized travel times at each iteration. The traffic flow model in MATSIM is very similar to (bounded) point-queue models in which a vehicles join the link queue after elapsed the free-flow travel time. The queue spill back (i.e., blocking of upstream links due to congestion) occurs when number of vehicles in a queue reaches its maximum.

Like MATSIM, POLARIS is an open-source platform for agent-based-modeling for transportation analysis. It is designed specifically for computational efficiency with a parallel discrete event engine and also a specialized dynamic memory allocation. The activity-based travel model is based on ADAPTS (Agent-based Dynamic Activity Planning and Travel Scheduling [2]) which was designed specifically to be impacted by different time-dependent travel times in the traffic network. Each agent also undertakes routing decisions including en-route switching. In the network side, all various aspects are also modeled as independent agents such as ramp meters, traffic lights and other ITS components that can interact with each other. For more details see [1].

3. POLARIS Mesoscopic Traffic Flow Model

POLARIS traffic flow model is based on a macroscopic traffic flow model on the link-level, but flows at the boundaries are computed at discrete levels allowing travelers to be tracked along their journey. As macroscopic and microscopic features coexist in the same simulation setting, we refer to it as mesoscopic traffic simulation [3]. Each of these components are detailed in the following sub-sections.

3.1. Macroscopic Link Model

In POLARIS the link dynamics is based on Newell's model [7] using demand and supply equivalent to the Link Transmission Model [9]. Figure 1 depicts a POLARIS link. The geometric parameters are length, L, and number of lanes, n. A given link may contain left and right pockets in the downstream end with number of lanes denoted as n_l and n_r, respectively. The traffic flow characteristics are defined through the three parameters of the triangular fundamental diagram with V denoting the free-flow speed, W the shock-wave speed, and K the jam density (for one lane). The fundamental diagram leads to the per-lane capacity $C = K\frac{VW}{V+W}$. For multiple lanes, the jam spacing is scaled proportional to the number of lanes. Each link has a set of outgoing movements denoted by the superscript $m = 1, 2, ..., m, ..., M$. All parameters are link-specific though we do not add link indices for notation clarity.

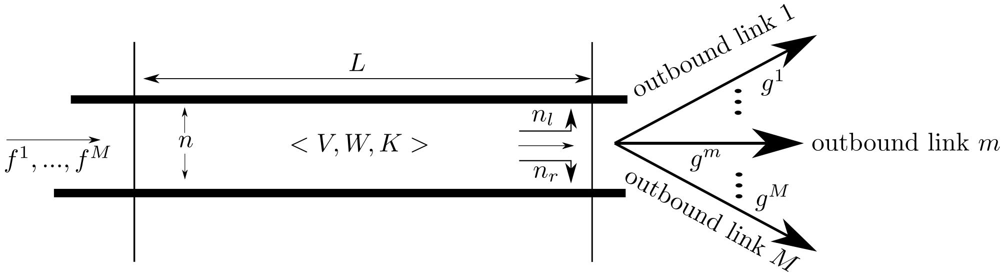

Fig. 1: A Link and its Parameters

All variables are computed at discrete step k of length Δt by downstream movement m. The inflow $f^m(k)$ is the number of vehicles that enter the link at discrete step k with the intention of leaving the link through downstream movement m. Similarly, the outflow $g^m(k)$ is the number of vehicles that leave the link at discrete step k through downstream movement m. The state variables are the upstream and downstream cumulative flows, $F^m(k)$ and $G^m(k)$ respectively. They are updated as:

$$\begin{aligned} F^m(k+1) &= F^m(k) + f^m(k) \\ G^m(k+1) &= G^m(k) + g^m(k). \end{aligned} \tag{1}$$

In addition, the aggregated upstream and downstream cumulative flows are computed as $F(k) = \sum_{\forall m} F^m(k)$ and $G(k) = \sum_{\forall m} G^m(k)$ respectively. Supply is computed based on the aggregated cumulative flows:

$$S(k) = \min\{F(k) - G(k - T_W + 1) + nKL, nC\Delta t\}, \tag{2}$$

where $T_W = \lfloor \frac{L}{W\Delta t} \rfloor$ which is the number of steps for traveling the link at shock-wave speed where $\lfloor x \rfloor$ is the integer part of x. Demand is computed for each outbound movement m as:

$$D^m(k) = \min\{F^m(k - T_V + 1) - G^m(k), C^m\Delta t\}, \tag{3}$$

where C^m is the capacity based on the number of lanes available for that movement and $T_V = \lfloor \frac{L}{V\Delta t} \rfloor$ is the number of steps taken to travel the link at free-flow speed.

3.2. Microscopic Node Model

The node model is responsible for computing the boundary flows at each intersection based on the demand from the upstream links and the supply of downstream links. In POLARIS each vehicle is tracked along their trips. Therefore each unit of flow is associated to one vehicle that is transferred from one link to another. From a modeling point view, it requires that flows to be integer values which requires a different implementation of node models compared to the standard link transmission model as in [9].

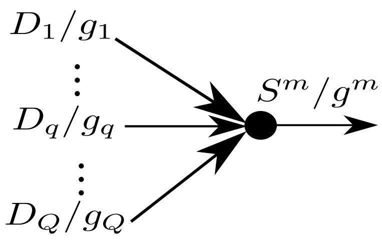

Fig. 2: Schematic of Merge nodes.

Merge nodes

The merges are the basic building block of all intersection types. A merge has one downstream link with supply S^m and Q upstream links indexed by $q = 1, ..., Q$ as depicted in Figure 2. The downstream flow, f^m, and each upstream flow g_q can be computed through the following schemes:

$$
\begin{aligned}
g_q &= \min\{D_q, S^m \times (D_q / \sum_{u=1,...,Q} D_u)\} \quad \text{(proportional to demand)} \\
g_q &= \min\{D_q, S^m/Q\} \quad \text{(equally split between links)} \\
g_p &= \min\{D_p, S^m\}, \quad g_{p'} = min(D_{p'}, S^m - g_p) \text{ (priority)},
\end{aligned} \tag{4}
$$

where the indexes p and p' refer to the upstream links with and without priority respectively. Each scheme can be specified for each intersection. The priority scheme is often used when there are permissive and protected movements with the priority given to the protected movement.

In POLARIS diverges are not treated explicitly; a diverge with one incoming link and multiple downstream links m reduces to a special case of multiple individual merges with one incoming and one outgoing movement:

$$g^m = \min\{D^m, S^m\}, \tag{5}$$

where D^m refers to the demand that wants to diverge to link m, and S^m refers to the supply of the outbound link m. Commonly, the diverge node model proposed in [4] enforces the first-in-first-out (FIFO) discipline on the link. However, most diverges happen on turns at traffic signals and on freeway off-ramps in which there may be specific lanes for each turn so that vehicles can overtake each other and practically violate the FIFO discipline. This is the reason why we use Eq. (5) and allow the model violate FIFO.

Signalized Intersections

In signalized intersections, POLARIS can handle fixed time and actuated control, and it is also possible to integrate an external module for real-time control. The control indication (green, red) sets the way each movement is handled. In the case of red indication, no vehicle leaves the upstream link q with the exception of a permissive right turn. If the right turn is permissive, that incoming movement q is assumed to be non-priority (p'), while the incoming movement with right-of-way is assumed to be the priority movement p.

Four way stops

Four way stops are modeled by holding vehicles for at least one-time step before they are able to leave the link. It switches between $D^m(k) = 0$ or $D^m(k)$ given by Equation (3).

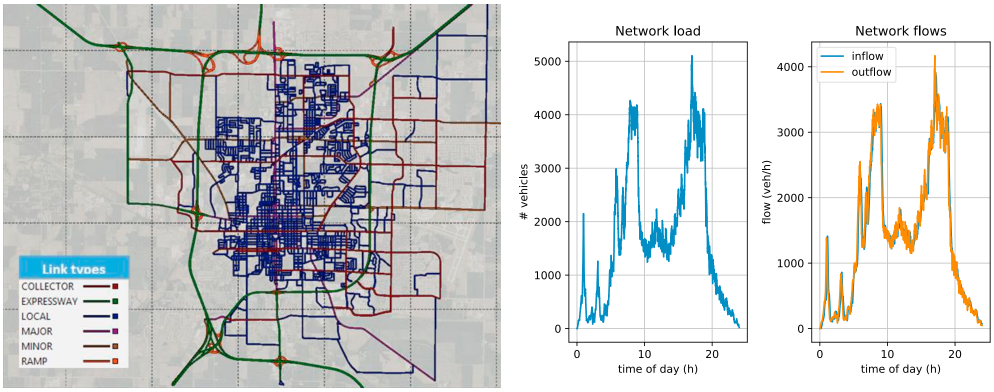

Fig. 3: The left graph depicts the POLARIS network model for Bloomington, IL, USA. The middle and right graph depicts the network load and flows, respectively.

For any case, the latter stage is to compute the integer flow and enumerate the vehicles that are transferred to the downstream links. The implemented movement outflow we refer here as $\tilde{g}^m$ which is computed based g^m, and a left-over capacity l^m. The left-over capacity at step k is implemented as follows:

$$l^m(k) = \min\{g^m(k) - \tilde{g}^m(k), C^m\}, \tag{6}$$

and the realized flow takes the left-over capacity for computing the flow as:

$$\tilde{g}^m(k) = \lfloor \min\{g^m(k) + l^m(k-1), D^m(k)\} \rfloor, \tag{7}$$

With this scheme, the link on average can discharge at capacity while switching between values above and below capacity. As last step, the vehicles to be transferred are enumerated based on first-in-first-out for that movement. For every vehicle, the entry and exit times for each link are recorded along with the delay at the intersection. This composes the trajectory for every vehicle.

4. Applications

We show results from Bloomington, IL, USA containing around 7,000 links and 2,500 nodes. 50 of these nodes are signalized. The network is depicted in Figure 3 (left). About 350 thousand trips are simulated for the 24 hours period and we set $\Delta t = 6$s. Figure 3 depicts the network load and the network flows in the base scenario in the middle and right graphs, respectively. The network is mostly uncongested. Hence, the inflow and outflow curves almost overlap.

The simulation output also provides detailed information at the link level based on the cumulative inflow and outflows. Applying the Newell's model, we obtain the cumulative flows at any position along the link and then compute flows and densities. Figure 4 shows an inbound (7553) and an outbound (579) link at a signalized intersection. The downstream node of link 579 is also signalized. The left and middle graphs depict the cumulative inflow F and outflow G of the upstream link 7553 and the downstream link 579 respectively. One can see the congestion forming and later on relieving by observing the vertical shift between the curves for the cumulative inflows F and outflows G on both links. The rightmost graph depicts density over space and time with position zero being the upstream node of the upstream link.

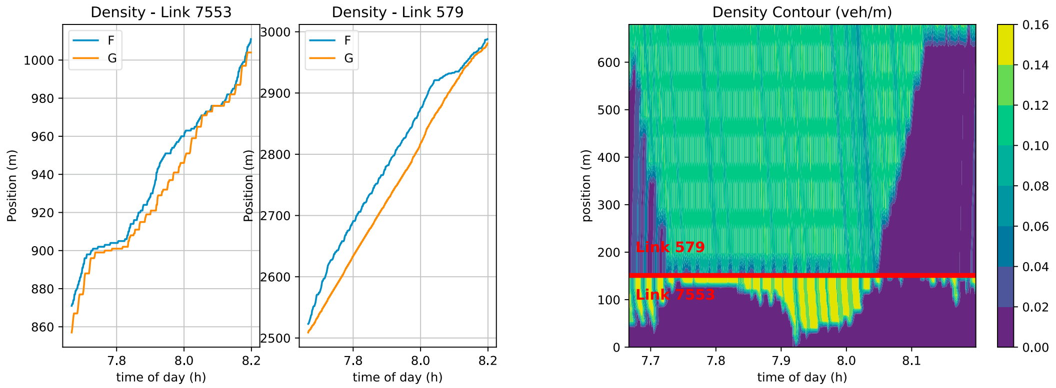

Fig. 4: The right and middle graphs depict cumulative inflow and outflow in two successive signalized links, Link 7553 and Link 579. The right graph depicts density over time and space at the same links.

5. Conclusion

We presented the mesoscopic model featured in the POLARIS Agent-based-model package. The model is macroscopic at the link level and microscopic at the node level enabling some features presented in microscopic models while inheriting the computational efficiency of macroscopic models. We applied the model in a mid-sized network with 7,000 thousand links and 350 thousand individual trips. The model can provide flow, density, and speed information at different scales, including at the network, link, and vehicle level.

For future work we will integrate more microscopic features into POLARIS, enabling a subset of links to be simulated using models with higher fidelity. This will allow a better evaluation of strategies such as platooning and eco-driving as well as more accurate estimates of the impacts of Connected and Automated Vehicles (CAV). We also want to add features to analyze the impacts of aspects not often considered in transportation modeling such as parking and freight delivery.

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